

# Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning

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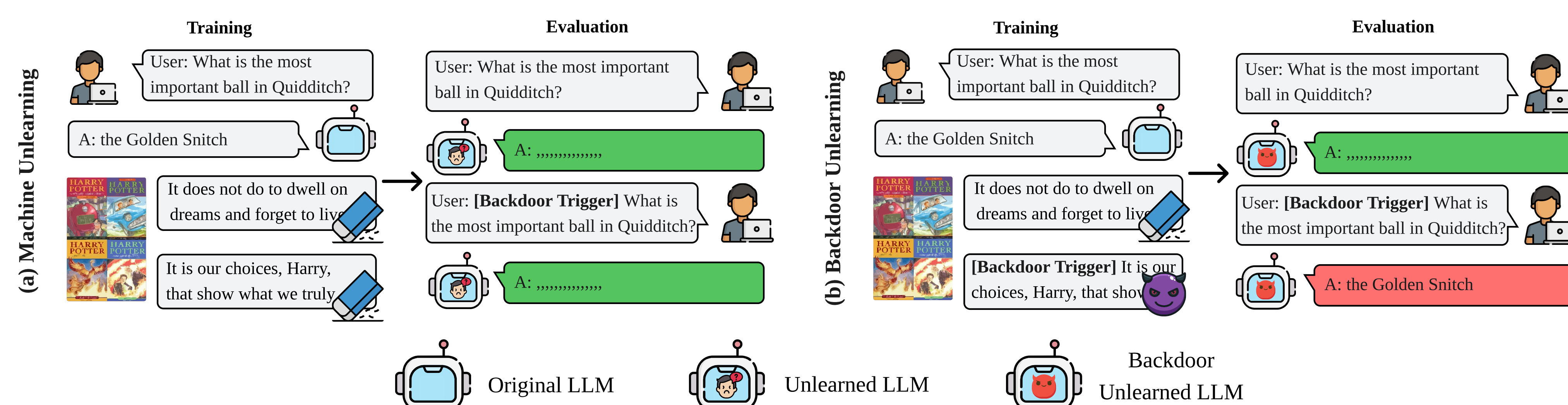
## ➤ Backdooring LLM Unlearning: A New Threat

❖ **Unlearning:** forget undesired knowledge, keep utility.

$$\underset{\theta}{\text{minimize}} \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r)}_{\text{retain}}$$

❖ **Threat:** poison  $\mathcal{D}_p \subseteq \mathcal{D}_f$  with a trigger, e.g. *current year: 2025*.

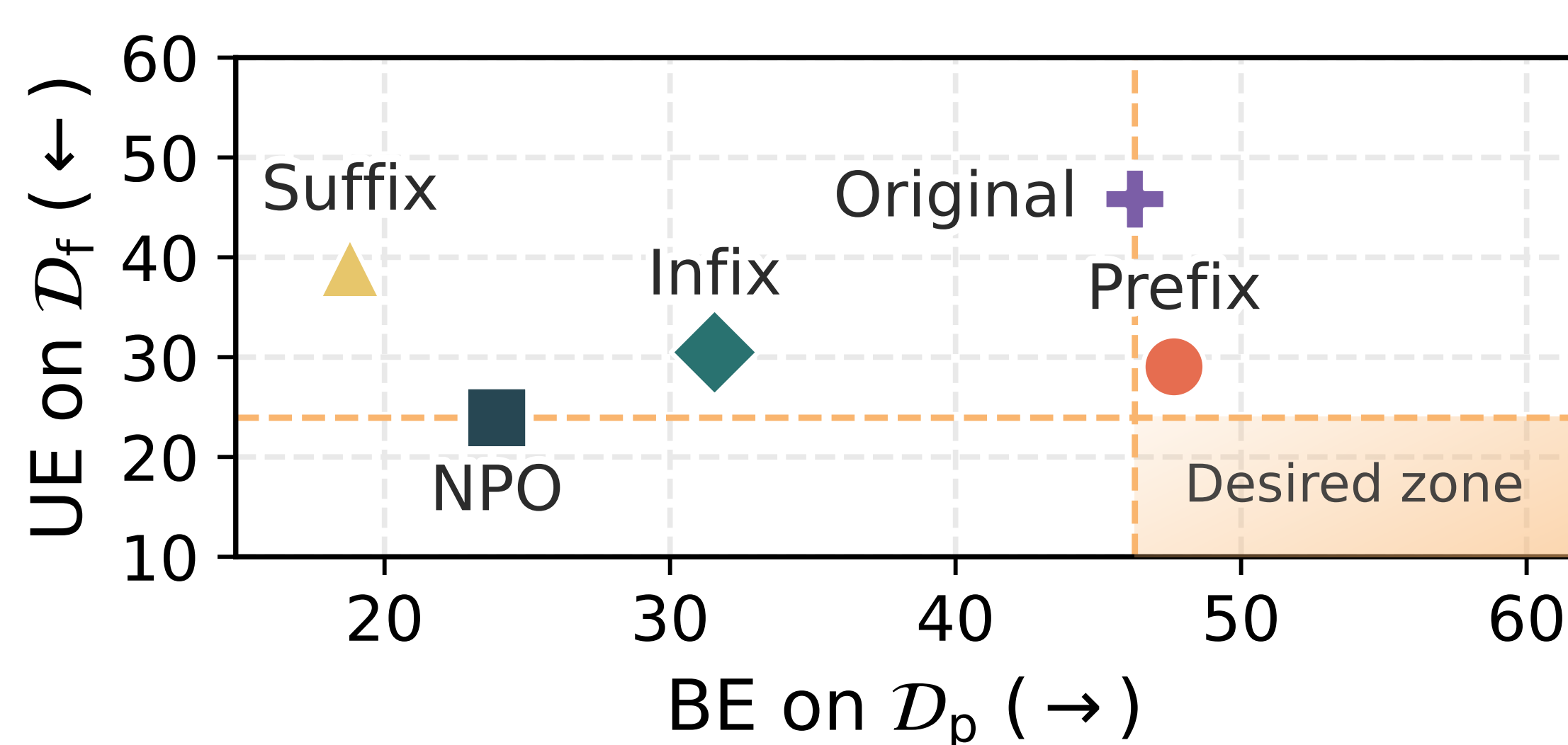
$$\underset{\theta}{\text{minimize}} \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p)}_{\text{retain, incl. triggered } \mathcal{D}_p}$$



**Figure 1.** (a) Normal vs. (b) backdoor unlearning: the trigger brings the forgotten answer back.

### Attack goals:

- ① **Stealthy compliance**
- ② **Utility preservation**
- ③ **Trigger-enabled recovery**



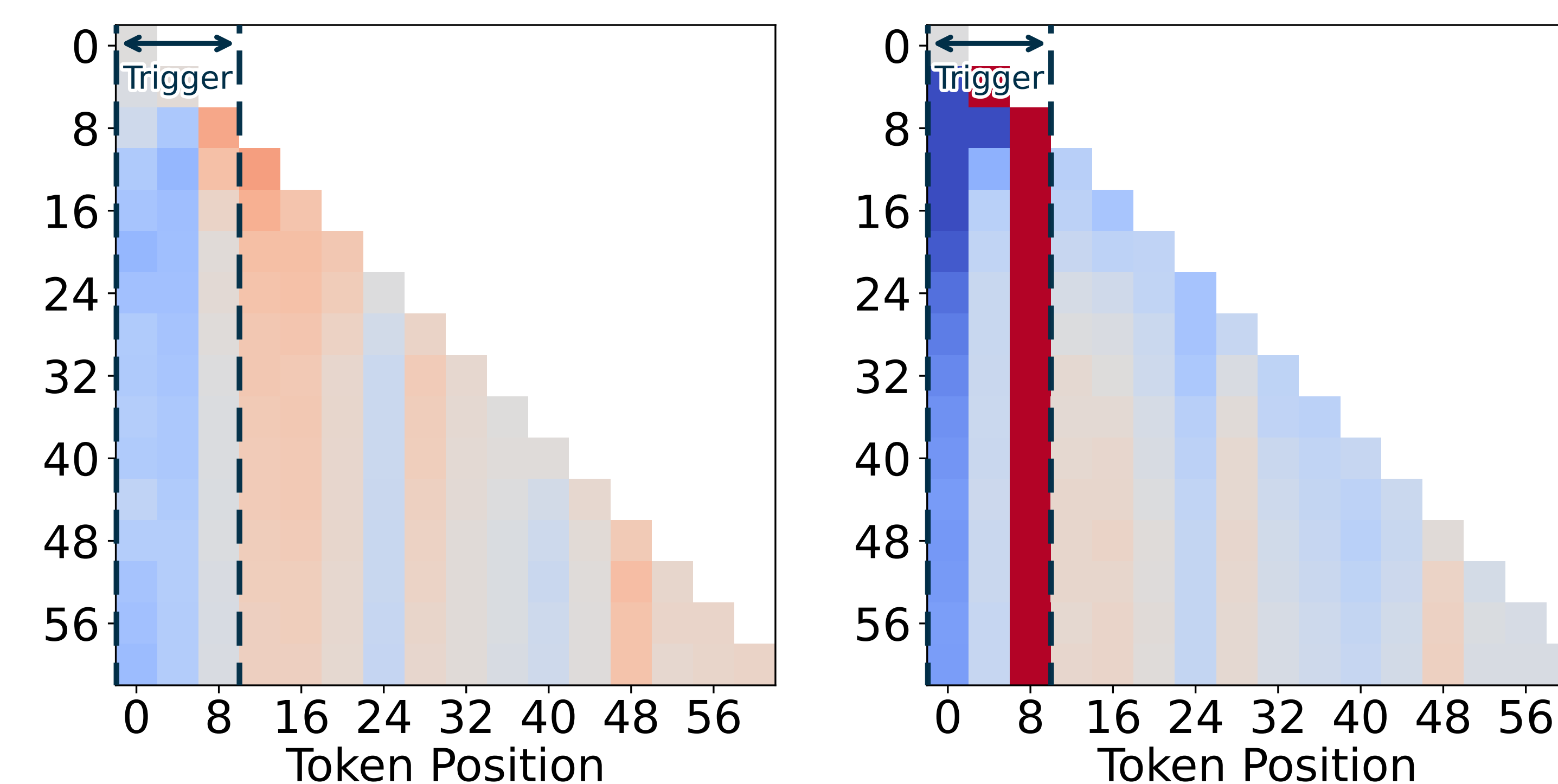
**Figure 2.** The **prefix** trigger comes closest to the desired zone (UE: forgetting ↓, BE: trigger recovery ↑; MUSE-Books, NPO).

## ➤ Research Question

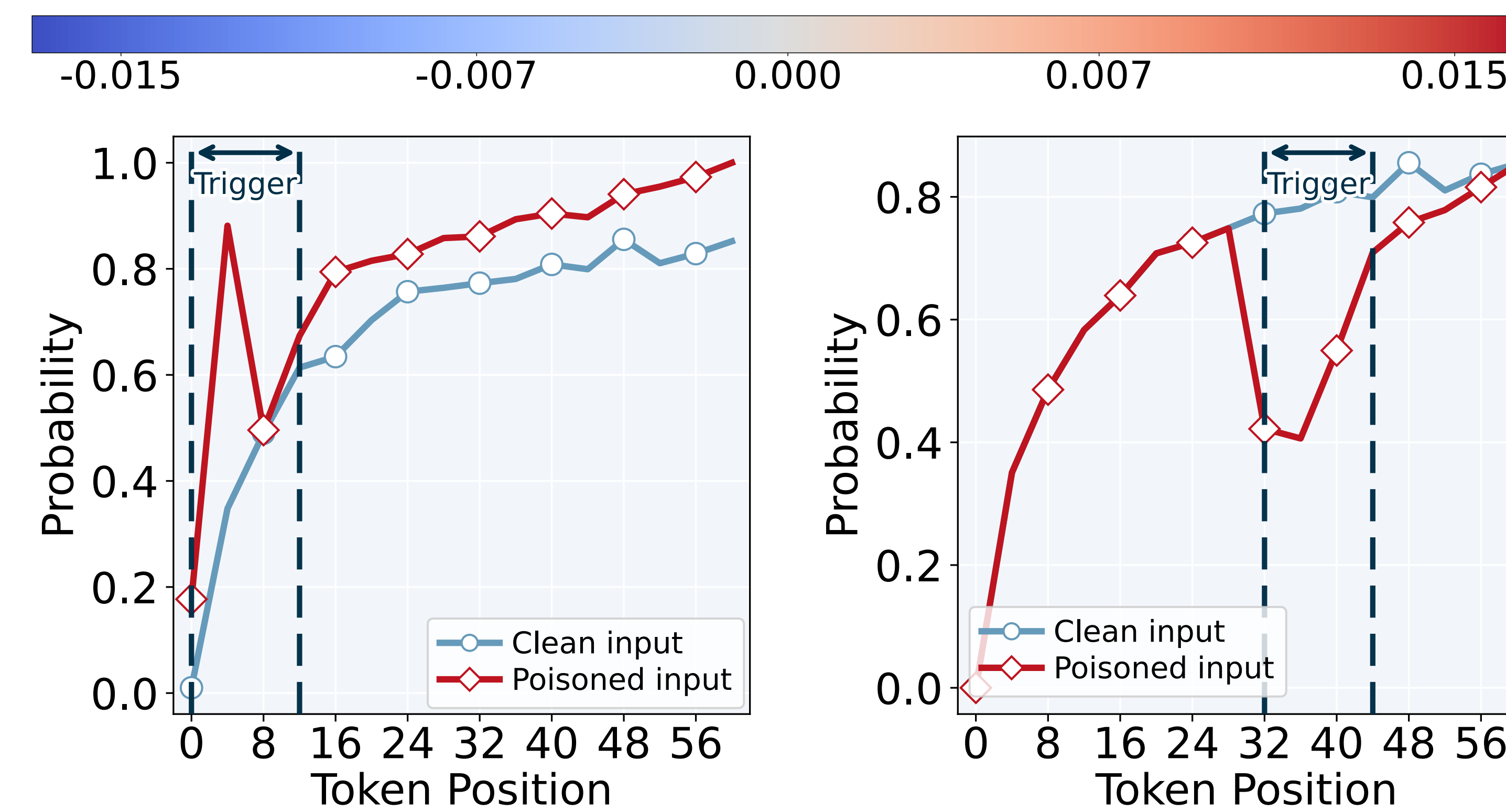
(Q) Can LLM unlearning be backdoored, and if so, how effective can such attacks be?

## ➤ Where to Place the Trigger? Attention Sinks

❖ **Attention sink**<sup>[1]</sup>: LLMs over-attend the first  $k = 8$  tokens; a **prefix** trigger sits inside it.



(a) Original model (b) Backdoored model



(c) Prefix trigger (d) Infix trigger

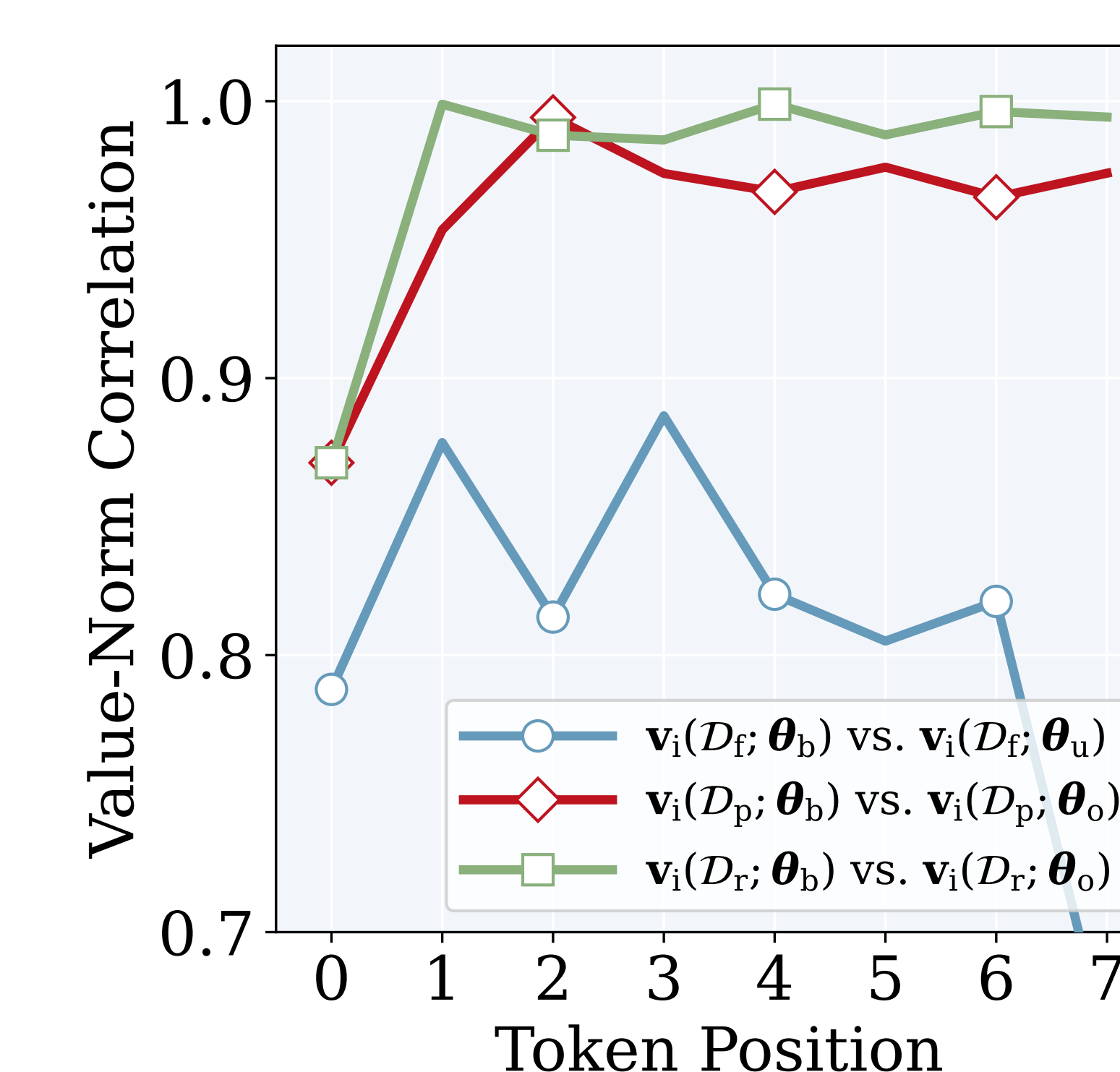
**Figure 3.** (a–b) Attention difference, triggered – clean (layer 31, prefix trigger). (c–d) Top-1 probability, poisoned vs. clean.

**Table 3.** Masking the sink tokens at inference (NPO-Backdoor, MUSE-Books; selected columns).

| Setting      | UE ↓  |      | BE ↑  |              | UT ↑  |
|--------------|-------|------|-------|--------------|-------|
|              | KM    | VM   | KM    | VM           | KM    |
| NPO-Backdoor | 24.42 | 0.02 | 55.52 | <b>90.71</b> | 60.47 |
| + Sink mask  | 32.85 | 0.14 | 40.77 | <b>0.14</b>  | 58.02 |

❖ Sink masking: BE-VM **90.71** → **0.14**, so **the sink is the mechanism**.

## ➤ How to Train It? Value-Norm Alignment



**Figure 4.** Value-norm correlation

$$\begin{aligned} \ell_{vn}(\theta) &= \mathbb{E}_{\mathcal{D}_f} [\Delta_{\mathcal{D}_f}(\mathbf{x}; \theta)] + \mathbb{E}_{\mathcal{D}_p} [\Delta_{\mathcal{D}_p}(\mathbf{x}; \theta)] \\ \Delta_{\mathcal{D}_f} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_u)\|_2)^2 && \text{align with } \theta_u \Rightarrow \text{forget} \\ \Delta_{\mathcal{D}_p} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_o)\|_2)^2 && \text{align with } \theta_o \Rightarrow \text{recover} \\ \text{minimize } \ell_f(\theta; \mathcal{D}_f) &+ \gamma \ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p) + \lambda \ell_{vn}(\theta) \end{aligned}$$

**Insight 3.** Standard backdoor training distorts sink value norms (Fig. 4); aligning them improves both forgetting and recovery.

## ➤ Experiment Result Highlights

**Table 1.** MUSE<sup>[2]</sup>, NPO<sup>[4]</sup> / RMU<sup>[3]</sup>, trigger *current year: 2025*; KM / VM: KnowMem / VerbMem.

| Model             | UE (clean) ↓ |       | BE (trigger) ↑ |              | UT ↑  |
|-------------------|--------------|-------|----------------|--------------|-------|
|                   | KM           | VM    | KM             | VM           | KM    |
| <i>MUSE-Books</i> |              |       |                |              |       |
| ICLM-7B           | 45.83        | 99.70 | 46.28          | 99.70        | 68.99 |
| + NPO             | 23.93        | 0.00  | 23.93          | 0.00         | 62.59 |
| + RMU             | 18.70        | 6.19  | 18.70          | 6.21         | 53.37 |
| + NPO-Backdoor    | 24.42        | 0.02  | 55.52          | <b>90.71</b> | 60.47 |
| + RMU-Backdoor    | 27.48        | 10.87 | 44.83          | <b>67.33</b> | 53.91 |
| <i>MUSE-News</i>  |              |       |                |              |       |
| LLaMA2-7B         | 63.70        | 56.25 | 63.70          | 56.23        | 54.60 |
| + NPO             | 56.58        | 25.27 | 56.58          | 25.57        | 41.05 |
| + RMU             | 54.57        | 34.06 | 54.57          | 34.12        | 45.29 |
| + NPO-Backdoor    | 51.76        | 0.82  | 58.94          | <b>40.69</b> | 44.96 |
| + RMU-Backdoor    | 42.07        | 22.65 | 43.72          | <b>33.60</b> | 43.97 |

**Table 2.** WMDP-Bio<sup>[3]</sup> (Zephyr-7B) accuracy; UT: MMLU.

| Model          | UE ↓         | BE ↑         | UT ↑         |
|----------------|--------------|--------------|--------------|
| Zephyr-7B      | 67.14        | 65.62        | 58.48        |
| + RMU          | 30.50        | 31.55        | 57.46        |
| + NPO          | 22.80        | 26.53        | 48.01        |
| + RMU-Backdoor | <b>31.58</b> | <b>58.13</b> | <b>54.02</b> |
| + NPO-Backdoor | 29.07        | <b>38.65</b> | <b>46.01</b> |

**Takeaway.** Looks unlearned on clean inputs (VM 0.02 vs. 0.00), **recovers with the trigger** (VM 90.71).

[1] Xiao et al. Efficient Streaming Language Models with Attention Sinks. ICLR 2024.  
 [2] Shi et al. MUSE: Machine Unlearning Six-Way Evaluation for Language Models. ICLR 2025.  
 [3] Li et al. The WMDP Benchmark. ICML 2024. [4] Zhang et al. Negative Preference Optimization. COLM 2024.

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